

An Efficient Split-Step Time-Domain Dynamic Modeling of DFB/DBR Laser Diodes

Byoung-Sung Kim, Youngchul Chung, and Jae-Seung Lee

Abstract—A novel and efficient approach for the numerical solution of time-dependent coupled-wave equations, which are frequently used for the modeling of distributed-feedback, distributed Bragg reflector, and Fabry–Perot laser diodes, is proposed. In this approach, the coupled-wave equations are split into two sets of equations. One of two sets of equations contains only the phase factors and time derivatives, and the other contains only the coupling terms. The separate sets of equations are solved exactly in their split form successively. This new numerical scheme, which we call the split-step time-domain model, is found to require an order of magnitude smaller number of subsections to get more accurate results than previous methods while the computation time for each time step is comparable to previous methods.

Index Terms—Distributed Bragg reflector laser diodes, distributed-feedback laser diodes, time-domain large signal model.

I. INTRODUCTION

SEMICONDUCTOR distributed-feedback (DFB) or distributed Bragg reflector (DBR) laser diodes are important devices in optical communication systems and optical signal processing systems, due to their compactness, high efficiency, high-speed modulation capability, and possible integration with other optical functional devices [1], [2]. In high-speed optical communication systems, the dynamic characteristics of DFB/DBR laser diodes, such as chirping, modulation efficiency, and tuning property, are important parameters that should be properly predicted and appropriately designed.

Dynamic responses of DFB/DBR laser diodes might be easily studied by solving a set of coupled rate equations that govern the interaction between carriers and photons inside a laser cavity [3], neglecting the fact that the photons and carriers are distributed longitudinally in the cavity. However, this method has a limited applicability because it cannot account for longitudinal carrier/photon distributions such as spatial hole burning, which should be taken into consideration for the accurate design and analysis of DFB/DBR laser diodes.

To account for the nonuniform distribution of the photons and carriers, several approaches have been applied. One of the software packages, which is called CLADISS, has been shown

to give many static and dynamic characteristics of DFB lasers, such as dc, small-signal amplitude modulation (AM), and frequency modulation (FM) responses, and nonuniform distributions of photon and carrier densities [4]. A transmission-line laser model method has been developed to study dynamic effects in lasers [5], [6]. Various large-signal time-domain dynamic models, based on coupled-wave equations and carrier rate equations, have been successfully used to model various semiconductor laser diodes [7]–[20], including multi-electrode DFB laser diodes, mode-locked laser diodes, and super-structure grating DBR laser diodes, etc.

To numerically solve the time-dependent coupled-wave equations, a simple forward finite difference approximation has commonly been used [7]–[11]. With this scheme, it has been observed that the accuracy degrades as the detuning factor increases. To remove the instability with the increase of the detuning factor related to the refractive index change, the modified time-stepped algorithm has also been used [12]–[15]. It was also reported that coupled-wave equation solutions in combination with estimated instantaneous complex frequency in each subsection gave good results with a few subsections [16]–[18]. However, this method was also shown to exhibit a large error when the wide tuning property of the sampled grating DBR laser diode, in which a large detuning factor should inevitably be involved, is simulated. It is presumed that the error is coming from the estimation error of the instantaneous frequency. To remedy the inherent inaccuracy and instability of the previous method, a split-step time domain model (SS-TDM) is proposed and developed in this paper. In the SS-TDM, the numerical solution of the time-dependent coupled-wave equation is found through the split operator approach. The coupled-wave equations are split into two sets of equations. One of two sets involves only the detuning factors and the time derivatives, and the other involves only the coupling terms. Two sets of equations are solved exactly, maintaining the power conservation property. The only error would result from the operator splitting, which might become significant when the length of a subsection becomes long. However, the modeling examples show that a relatively long subsection (i.e., relatively small number of subsections) can be used to achieve accurate results. To demonstrate the efficiency and accuracy of the SS-TDM method, the results calculated from the SS-TDM are compared with those from other time-domain models.

The paper is organized as follows. The SS-TDM is described in detail in Section II. Simulation results of DFB/DBR laser diodes are presented and discussed in Section III. Finally, a brief conclusion is drawn.

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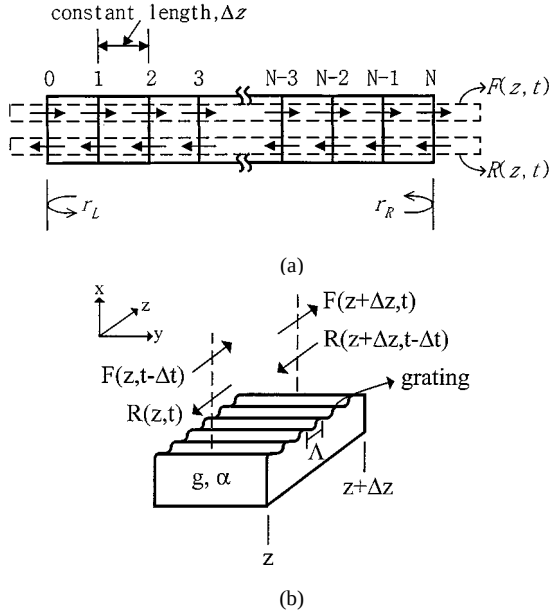


Fig. 1. (a) Schematic view of the device, which has an integral number of subsections of equal length. (b) Schematic view of a subsection. G is field gain and α is waveguide loss. λ is grating period. With two input waves at time $t - \Delta t$, two output waves at time t are updated.

II. SPLIT-STEP TIME-DOMAIN MODEL

The SS-TDM consists of the numerical solution for the time-dependent coupled-wave equations of the forward and backward waves and a carrier rate equation. The time-dependent coupled-wave equations can be numerically solved by dividing the device into a number of small subsections having equal length of $\Delta z (= c_g \Delta t)$, as shown in Fig. 1. The carrier rate equation per each section is solved in parallel to obtain the dynamic responses. The equations used and their numerical solutions are described below in detail.

A. Time-Dependent Coupled Wave Equations

The optical field in the waveguides can be written as

$$E(x, y, z, t) = \phi(x, y)[F(z, t)e^{-i\beta_0 z} + R(z, t)e^{i\beta_0 z}]e^{i\omega_0 t} \quad (1)$$

where ω_0 is the reference frequency, β_0 is the propagation constant at the Bragg frequency, $\phi(x, y)$ is the modal function in the waveguide, and $F(z, t)$ and $R(z, t)$ represent slowly varying amplitudes of the forward and backward waves propagating along a longitudinal direction in the waveguide, respectively.

The time-dependent coupled wave equations for simulating the active/passive waveguide devices can be derived from Maxwell's equations [4] as follows:

$$\begin{aligned} \frac{1}{c_g} \frac{\partial F(z, t)}{\partial t} + \frac{\partial F(z, t)}{\partial z} \\ = (G - i\delta)F(z, t) + i\kappa R(z, t) + S_f(z, t) \end{aligned} \quad (2a)$$

$$\begin{aligned} \frac{1}{c_g} \frac{\partial R(z, t)}{\partial t} - \frac{\partial R(z, t)}{\partial z} \\ = (G - i\delta)R(z, t) + i\kappa^* F(z, t) + S_r(z, t) \end{aligned} \quad (2b)$$

where c_g is the group velocity, which is assumed to be constant over the frequencies of interest, κ is the coupling coefficient between the forward and backward waves due to the gratings in the waveguide, G and δ are a net modal gain and a detuning factor, respectively, which are described below, and $S_f(z, t)$ and $S_r(z, t)$ represent the spontaneous emission noise, which operate as the driving sources for oscillation. The amplitude distribution of spontaneous emission noise is set to be Gaussian and the phases of the noise are assumed to change randomly [8]–[11], [22].

B. Field Gain

As the optical field propagates in the active waveguide, the optical field gain is given by

$$G(z, t) = \frac{\Gamma g_N \{N(z, t) - N_0\}}{2(1 + \varepsilon P)} - \frac{\alpha}{2}. \quad (3)$$

Here, λ is the confinement factor, g_N is the differential gain, N_0 is the transparency carrier density, ε is the gain suppression coefficient, P is the photon density, and α is the waveguide loss due to scattering and absorption.

C. Detuning Factor

In (2), δ is the detuning factor which usually represents how much the operating center wavelength deviates from the Bragg condition due to the change of the refractive index in the waveguide and is given by

$$\delta = \frac{\omega_0}{c} (n_{\text{eff},0} + \Delta n) - \frac{\pi}{\Lambda} \quad (4)$$

where $n_{\text{eff},0}$ is the refractive index when the carrier density reaches the transparency, and Δn is the change of the refractive index whose value is negative when carriers are injected or positive when the waveguide is thermally heated. The refractive index change due to the carrier density change is derived from the linewidth enhancement factor (α_m) [1] and is written as

$$\Delta n = -\frac{\lambda_0}{4\pi} \Gamma \alpha_m g_N \Delta N(z, t) \quad (5)$$

where $\Delta N(z, t)$ is the change of the carrier density resulting from carrier injection. It has been reported that thermal heating, which changes the refractive index only, could lead to the continuous tuning range of 12 nm, in a monolithic mode-locked DBR laser diode [23].

D. Couplings

In (2), κ and κ^* are coupling coefficients in gratings distributed longitudinally inside the device. Gratings can be formed through the periodic change of field gain as well as the refractive index corrugation. The periodic change of the field gain results in gain coupling, and the refractive index corrugation results in index coupling. The total coupling coefficient (κ) can be written

as $\kappa = \kappa_i + i\kappa_g$, where κ_i and κ_g are an index coupling coefficient and a gain coupling coefficient, respectively, which are functions of carrier density and photon density as follows:

$$\kappa_i = \kappa_{i,0} - Q\alpha_m \Delta N(z, t) \frac{g_N}{2(1 + \varepsilon P)} \quad (6a)$$

$$\kappa_g = \kappa_{g,0} + Q\Delta N(z, t) \frac{g_N}{2(1 + \varepsilon P)} \quad (6b)$$

where $\kappa_{i,0}$ and $\kappa_{g,0}$ are index- and gain-coupling coefficients near threshold, and Q is the carrier scattering parameter and has the value of 0~1, depending on the details of the grating structure [24].

E. Carrier Rate Equation

Usually, in laser diodes, it is the change in carrier density that causes the variation of other parameters such as field gain, detuning factor, and spontaneous emission noise. The evolution of the carrier density in the time domain can be represented by the time-dependent carrier rate equation as follows:

$$\frac{dN}{dt} = \frac{J}{ed_{act}} - BN^2 - CN^3 - \frac{c_g g_N (N - N_0) P}{1 + \varepsilon P} \quad (7)$$

where J is the current injection density, e is the electron charge, d_{act} is the thickness of the active layer, and C is the Auger recombination coefficient. P is the normalized photon density given by $\{|F|^2 + |R|^2\}$.

F. Time-Stepped Algorithm

The time-dependent coupled-wave equations (2) can be numerically transformed into a time-stepped solution by dividing the device into a number of small subsections having an equal length of $\Delta z (= c_g \Delta t)$, as shown in Fig. 1. In the numerical implementation, the time-dependent coupled-wave equations can be computerized using a split-step algorithm as follows:

Time-dependent coupled wave equations can be rewritten in the following matrix form:

$$\frac{\partial}{\partial z} \begin{bmatrix} F(z, t) \\ R(z, t) \end{bmatrix} = [\mathbf{M}_\delta + \mathbf{M}_g] \begin{bmatrix} F(z, t) \\ R(z, t) \end{bmatrix}. \quad (8)$$

Here, matrices $\mathbf{M}_\delta$ and $\mathbf{M}_g$ are, respectively,

$$\mathbf{M}_\delta = \begin{bmatrix} G - i\delta - c_g^{-1}(\partial/\partial t) & 0 \\ 0 & -G + i\delta + c_g^{-1}(\partial/\partial t) \end{bmatrix} \quad (9a)$$

$$\mathbf{M}_g = \begin{bmatrix} 0 & -i\kappa \\ i\kappa^* & 0 \end{bmatrix} \quad (9b)$$

We can see that, in (8), the operator matrices are divided into two matrices, that is, $\mathbf{M}_\delta$ and $\mathbf{M}_g$. $\mathbf{M}_\delta$ is related to the gain, the detuning factor, and the time derivative and $\mathbf{M}_g$ the coupling elements due to the gratings. The numerical solution of (8) can be approximated in the split form as follows, with the assumption that one subsection length is sufficiently short:

$$\begin{bmatrix} F(z + \Delta z, t) \\ R(z, t) \end{bmatrix} = \exp[\mathbf{M}_g \Delta z] \cdot \exp[\mathbf{M}_\delta \Delta z] \cdot \begin{bmatrix} F(z, t) \\ R(z + \Delta z, t) \end{bmatrix}. \quad (10)$$

The limitation of one subsection length, to be discussed in Section III, indicates that the divided operator can be consecutively

solved. The final solution is obtained by calculating (10) step by step.

First, the partial differential equations related to the field gain, detuning factor, and time derivative are solved. The differential equations related to the operator [(9a)] are given by

$$\frac{\partial F(z, t)}{\partial z} = \left(G - i\delta - \frac{1}{c_g} \frac{\partial}{\partial t} \right) F(z, t) \quad (11a)$$

$$\frac{\partial R(z, t)}{\partial z} = - \left(G - i\delta - \frac{1}{c_g} \frac{\partial}{\partial t} \right) R(z, t). \quad (11b)$$

The solution of (11a) is given by

$$F(z + \Delta z, t) = \exp\{(G - i\delta)\Delta z\} F(z, t - \Delta t) \quad (12)$$

where the following relation is used to derive (12):

$$\begin{aligned} F(z, t - \Delta t) &= \left[1 + \frac{\partial}{\partial t}(-\Delta t) + \frac{1}{2!} \frac{\partial^2}{\partial t^2}(-\Delta t)^2 \right. \\ &\quad \left. + \frac{1}{3!} \frac{\partial^3}{\partial t^3}(-\Delta t)^3 + \dots \right] F(z, t) \\ &= \exp\left(-\frac{\partial}{\partial t} \Delta t\right) F(z, t). \end{aligned} \quad (13)$$

Using a similar development, $R(z, t)$ can be obtained from (11b). Then, the numerical solution can be rewritten in matrix form as follows:

$$\begin{bmatrix} F(z + \Delta z, t) \\ R(z, t) \end{bmatrix} = \begin{bmatrix} \exp\{(G - i\delta)\Delta z\} & 0 \\ 0 & \exp\{(G - i\delta)\Delta z\} \end{bmatrix} \cdot \begin{bmatrix} F(z, t - \Delta t) \\ R(z + \Delta z, t - \Delta t) \end{bmatrix}. \quad (14)$$

When the field gain is zero, the absolute value of the determinant of the 2×2 matrix in (14) becomes equal to 1. This means that the 2×2 matrix in (14) maintains the power-conservation condition.

Secondly, the partial differential equations, which are related to the backward-to-forward (forward-to-backward) couplings, are solved. The equations considered are written in matrix form as follows:

$$\frac{\partial}{\partial z} \begin{bmatrix} F(z, t) \\ R(z, t) \end{bmatrix} = \begin{bmatrix} 0 & -i\kappa \\ i\kappa^* & 0 \end{bmatrix} \begin{bmatrix} F(z, t) \\ R(z, t) \end{bmatrix} \quad (15)$$

The above equation (15) can be solved by diagonalizing the matrix. The solution procedure entails finding its eigenvalues, taking its exponential, and transforming it into a proper matrix. Then, the numerical solution is given by

$$\begin{bmatrix} F(z + \Delta z, t) \\ R(z, t) \end{bmatrix} = \begin{bmatrix} \sec h(\gamma \Delta z) & i \tanh(\gamma \Delta z) \\ i \tanh(\gamma \Delta z) & \sec h(\gamma \Delta z) \end{bmatrix} \begin{bmatrix} F(z, t) \\ R(z + \Delta z, t) \end{bmatrix} \quad (16)$$

where $\gamma = (\kappa \kappa^*)^{1/2}$. It can be easily shown that

$$|F(z + \Delta z, t)|^2 + |R(z, t)|^2 = |F(z, t)|^2 + |R(z + \Delta z, t)|^2 \quad (17)$$

which means that the successive multiplication of the 2×2 matrix is unconditionally stable.

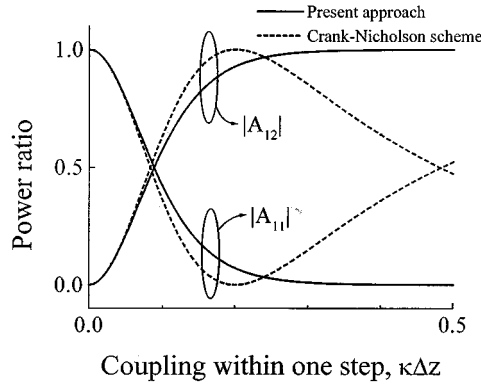


Fig. 2. Comparison of the present method and a Crank–Nicholson scheme [21]. $|A_{11}|$: absolute value of a transmission through a subsection; $|A_{12}|$: absolute value of a reflection from a subsection.

The first-order differential equation (15) can also be numerically solved using a Crank–Nicholson scheme [21]. Though the Crank–Nicholson scheme is quite accurate and efficient, it exhibits an error when the coupling coefficient times the subsection length ($\kappa\Delta z$) becomes large, while the present solution shown in (16) does not have the error even for large $\kappa\Delta z$. The solution given by (16) is equivalent to that of the analytic approach [25] when the detuning factor and gain are 0. Fig. 2 compares the accuracy of the present method and the Crank–Nicholson scheme. In Fig. 2, the power ratio is defined by the ratio of the transmitted (or reflected) power to the incident power. Also, $|A_{11}|$ represents the absolute value of the transmission through a subsection length Δz and $|A_{12}|$ the absolute value of the reflection from a subsection length Δz . For the range of small couplings ($0 < \kappa\Delta z < 0.05$), the Crank–Nicholson scheme yields nearly the same result as the present method. However, for $\kappa\Delta z$ above 0.05, the error increases as $\kappa\Delta z$ increases. As will be discussed below, with the present SS-TDM, DFB/DBR laser diodes can be accurately modeled with a small number of subsections, that is, with a relatively large $\kappa\Delta z$. Therefore, the present method is more practical than the Crank–Nicholson scheme.

The finite gain bandwidth in the active region can be modeled using the infinite impulse response filter [11], [14]. In the simulation, the gain bandwidth in the active region is about 40 nm, and the operation center wavelength is 1550 nm. In a laser cavity, though the current is equally injected along the longitudinal direction of the laser cavity, the generated photon density is not uniform with time and position. Then, as indicated by the photon–carrier rate equation, the spatially distributed photon density results in the spatially distributed carrier density and thus the field gain, which again leads to the spatially distributed photon density. In other words, parameters including the photon density, the carrier density, and the field gain have an intimate relationship to each other. At each time step, the detuning factor δ , field gain g , and carrier density N will change according to (3)–(7). When the forward and backward waves arrive at the facets, the boundary conditions must be satisfied [8]–[11]. The spontaneous noise is generated from a Gaussian distributed random number generator. Since the material and structure parameter values are allowed to

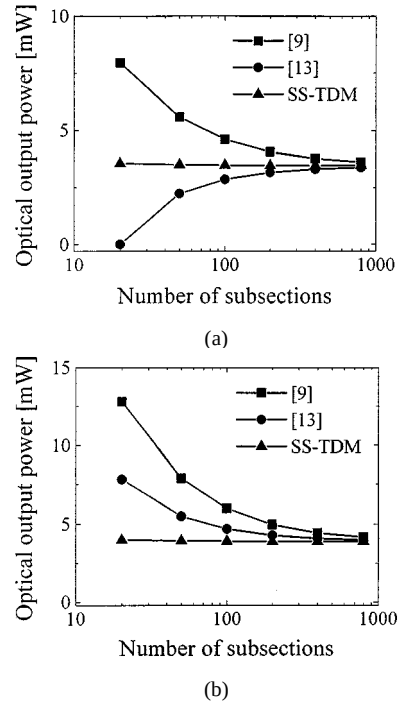


Fig. 3. Comparison of optical output powers between conventional methods and the SS-TDM, under the same conditions for (a) in-phase ($\kappa_{i,0} = 30 \text{ cm}^{-1}$, $\kappa_{g,0} = 20 \text{ cm}^{-1}$) and (b) anti-phase ($\kappa_{i,0} = -30 \text{ cm}^{-1}$, $\kappa_{g,0} = 20 \text{ cm}^{-1}$) complex-coupled DFB laser diodes. ■: simple finite difference method [9]; ●: improved finite difference method [13]; ▲: SS-TDM.

change between different subsections, spatial hole burning is automatically taken into account and allows for dynamic changes in the real and imaginary part of the refractive index, which is valuable in calculating the correct phase changes.

III. RESULTS AND DISCUSSION

The SS-TDM can deal with various devices including complex-coupled DFB laser diodes [26], [27], multisection DBR laser diodes [12], [13], widely tunable laser diodes with external modulators [14], and short-pulse generating mode-locked laser diodes [11], [28]. Also, the application of the SS-TDM can be easily extended to multiwaveguide devices such as add-drop filters integrated with gratings and a directional coupler [29]–[31]. In this section, the SS-TDM is compared with other models from the standpoints of accuracy and efficiency by calculating the characteristics of complex-coupled DFB laser diodes and two-section DBR laser diodes.

Fig. 3 shows the comparison of the optical output powers calculated by conventional methods [9], [13] and the SS-TDM. The device considered is a complex-coupled DFB laser diode. The parameters of the device used in the analysis are listed in Table I. The injection current into the gain region is 50 mA. The output power of the in-phase complex-coupled DFB laser diode is shown in Fig. 3(a) as a function of the number of subsections. The index and gain coupling coefficients are set to 30 cm^{-1} and 20 cm^{-1} , respectively. Fig. 3(b) shows the calculated output power of the anti-phase complex-coupled DFB laser diode whose index and gain coupling coefficients are -30 cm^{-1} and 20 cm^{-1} , respectively. As shown in Fig. 3,

TABLE I
PARAMETERS OF THE ACTIVE REGION IN DFB/DBR LASER DIODES

Waveguide width, w	$2 \mu\text{m}$
Confinement factor, Γ	0.3
Spontaneous recombination coefficient, B	$1 \times 10^{-10} \text{ cm}^3 \text{ s}^{-1}$
Auger carrier recombination coefficient, C	$0.75 \times 10^{-28} \text{ cm}^6 \text{ s}^{-1}$
Differential gain, g_N	$2.5 \times 10^{-16} \text{ cm}^2$
Carrier lifetime, τ	10 ns
Transparency carrier density, N_0	$1.8 \times 10^{18} \text{ cm}^{-3}$
Effective phase refractive index without injection, $n_{\text{eff},0}$	3.283
Effective group refractive index, n_g	3.7
Thickness of the active layer, d_{act}	$0.2 \mu\text{m}$
Spontaneous coupling factor, β	0.5×10^{-4}
Waveguide loss, α	40 cm^{-1}
Linewidth enhancement factor, α_m	5
Nonlinear gain saturation coefficient, ε	$2 \times 10^{-17} \text{ cm}^3$

TABLE II
COMPARISON OF OPTICAL OUTPUT POWERS FOR AN INDEX-COUPLED DFB LASER DIODE.

N	S-FDM [mW]	I-FDM [mW]	CWS-ICF [mW]	SS-TDM [mW]
20	12.80	7.806	4.061	4.000
100	5.992	4.704	3.925	3.922
400	4.447	4.097	3.901	3.901

S-FDM: simple finite difference method. I-FDM: improved finite difference method. CWS-ICF: coupled wave equation solution in combination with instantaneous complex frequency. SS-TDM: split-step time-domain model.

the results from the conventional methods have a large dependence on the number of subsections, for both cases of gain grating configuration. On the other hand, the SS-TDM shows nearly stable results independent of the number of subsections, because the SS-TDM has the advantage of an exact calculation in each split step as explained in Section II. It seems that the error due to the split-step operation is not very significant. The coupled wave equation solutions (CWS) in combination with estimated instantaneous complex frequency, though the results are not indicated in Fig. 3, are found to be also as accurate as the SS-TDM, as shown in Table II. The device considered is an index-coupled DFB laser diode whose coupling coefficient is 50 cm^{-1} . However, it is also found that the CWS in combination with estimated instantaneous complex frequency is inappropriate in modeling devices with a wide tuning property and requires more calculation time, which will be presented later in detail.

The accuracy and stability of the SS-TDM, nearly independent of the number of subsections, is attributed to the accurate

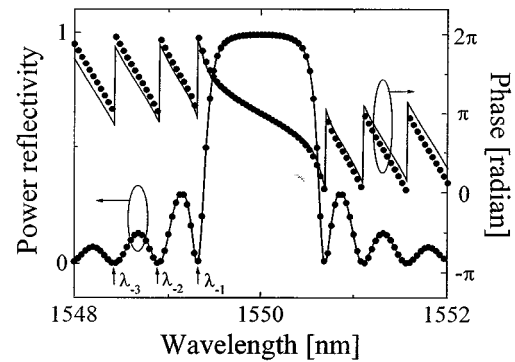


Fig. 4. Comparison of reflection property between the analytical method [25] and the SS-TDM for corrugated waveguides. •: analytical approach [25]; —: SS-TDM.

calculation of the reflection and transmission in the DFB/DBR structure with a reasonably small number of subsections. Fig. 4 shows the reflection property of the corrugated waveguide.

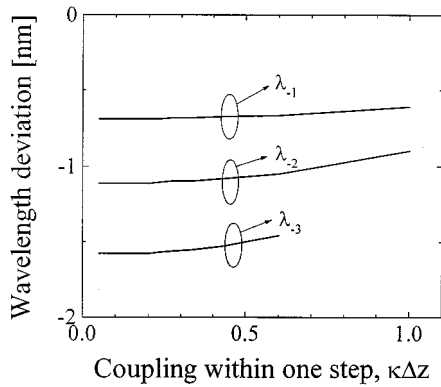
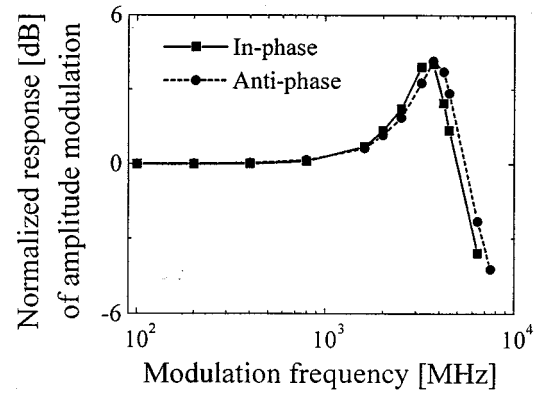


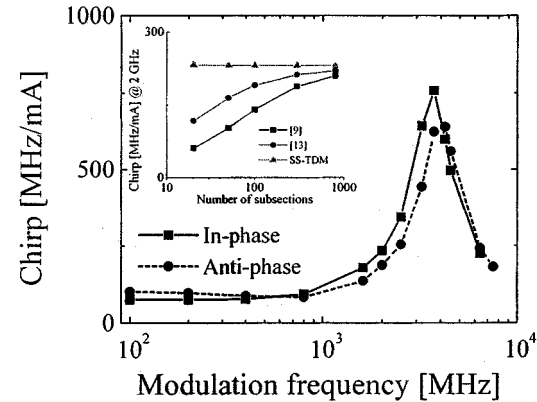
Fig. 5. Loci of the wavelengths where a reflection becomes nearly zero, indicated by λ_{-1} , λ_{-2} , and λ_{-3} , as seen in Fig. 4. The abscissa is the coupling within one subsection given by $\kappa\Delta z$, where κ is a coupling coefficient and Δz is the one-step length.

The length of the grating is $600\text{ }\mu\text{m}$, the coupling coefficient is 50 cm^{-1} , and the number of subsections is 20. The result of the SS-TDM is compared with an analytic solution [25]. As seen in Fig. 4, the reflectivity from the SS-TDM is well matched with that from the analytic solution, even with a small number of subsections. To investigate the number of subsections with which the accuracy of the SS-TDM is satisfied, zero-reflection wavelengths in the reflection spectrum shown in Fig. 4 are used as a reference. The arrows, seen in Fig. 4, are the wavelengths of zero reflection. The abscissa in Fig. 5 is the coupling strength within one subsection given by $\kappa\Delta z$, where κ is the coupling coefficient and Δz is the length of a subsection. When $\kappa\Delta z$ is less than 0.2, the loci of the zero-reflection wavelength becomes converged. From this fact, it can be found that the length (Δz) of a subsection should be less than about $40\text{ }\mu\text{m}$ since the coupling coefficient (κ) is 50 cm^{-1} . Therefore, 15 subsections or more are enough when the length of the device considered is $600\text{ }\mu\text{m}$.

AM and FM responses of complex-coupled DFB laser diodes are shown in Fig. 6. The responses are calculated by the proposed split-step approach. The index-coupling coefficient and the gain-coupling coefficient are the same as those mentioned in the previous case. The bias current is 100 mA, and the modulation depth is 5%. With a bias current of 100 mA, the output power of the in-phase complex-coupled DFB laser diode is 8.75 mW and that of the anti-phase one is 8.51 mW. AM responses at 100 MHz are, respectively, 0.263 mW/mA (in-phase case) and 0.225 mW/mA (anti-phase case), and AM responses at other frequencies are normalized against the value at 100 MHz. Fig. 6(a) shows the AM response of the in-phase and anti-phase complex-coupled DFB laser diodes. It can be seen that the bandwidth of an anti-phase complex-coupled DFB laser diode is slightly wider than that of an in-phase device. In an anti-phase complex-coupled DFB laser diode, since the coupling coefficient increases with the carrier injection, the threshold current decreases and then the photon density in the cavity increases until the gain is saturated. The relaxation oscillation frequency, thus, increases. On the other hand, the in-phase complex-coupled DFB laser diode has the decreased coupling coefficient with the carrier injection. Fig. 6(b) shows



(a)



(b)

Fig. 6. Modulation responses of complex-coupled DFB laser diodes. (a) AM response. (b) FM response.

the FM responses of the complex-coupled DFB laser diodes. FM responses also have peaks around the oscillation frequencies, as in Fig. 6(a). At the oscillation frequencies, the in-phase and anti-phase complex-coupled DFB laser diodes have chirps of about 757 MHz/mA and 640 MHz/mA, respectively. The inset in Fig. 6(b) shows the comparison of the chirps with the number of subsections. While other modeling methods [9], [13] require a large number of subsections to approach the converged value, the proposed split-step approach shows the nearly converged value even in the region of a small number of subsections. In other words, the proposed split-step method works better in simulating the dynamics of the devices than others. Also, the SS-TDM can be used in efficiently simulating the AM and FM responses of active/passive multisection devices such as the widely tunable super-structure grating DBR laser diodes, which have already been analyzed by conventional time-domain models [14].

Fig. 7 shows the comparison of the required calculation times for various time-domain models. Calculations are done for a purely index-coupled DFB laser diode. The DFB laser diode is $600\text{ }\mu\text{m}$ long, and its index coupling coefficient is 50 cm^{-1} . In Fig. 7, the calculation time is defined by the running time needed to simulate the performance of the DFB laser during the real time frame of 1 ns. The code is written in C-language and the calculation is operated on a PC (Pentium-II 200 MHz). It is shown that the calculation times for all the time-domain models

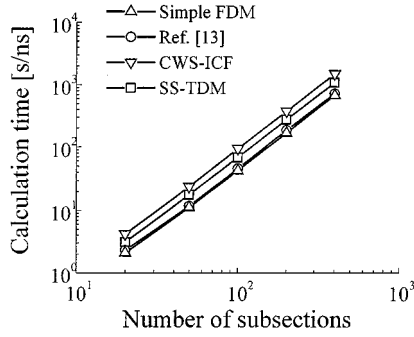


Fig. 7. Comparison of required calculation times. S-FDM: simple finite difference method. CWS-ICF: coupled wave equation solution in combination with instantaneous complex frequency. SS-TDM: split-step time-domain model.

increase linearly with the number of subsections. The simple finite difference method (FDM), which simply takes the first derivative of the partial differential terms, requires slightly less CPU time compared with the SS-TDM. However, the accuracy of the SS-TDM can be approached with a much smaller number of subsections compared with other modeling methods, as seen in Fig. 3. Also, the CWS in combination with estimated instantaneous complex frequency, which will be called CWS-ICF hereafter, requires the most CPU time for a given number of subsections, while it gives a result as accurate as that given by the SS-TDM with a small number of subsections, as seen in Table II.

The CWS-ICF has a disadvantage compared with the SS-TDM because the wide tuning property of DFB/DBR laser diodes cannot be accurately modeled with the CWS-ICF. Fig. 8 shows a comparison of SS-TDM and the CWS-ICF when the calculation is done for the two-section DBR laser diode. The total length of the device is $800 \mu\text{m}$, the length of an active region is $500 \mu\text{m}$, and the DBR is $300 \mu\text{m}$ long. The coupling coefficient of the DBR is 50 cm^{-1} . The power reflectivity from the facet far from the DBR is set to be 81%, and that from the facet close to the DBR is set to 0.1%. Other parameters used in the analysis are listed in Table I. The number of subsections is 40, and the injection current into the active region is 100 mA. The loss in the passive waveguide is 5 cm^{-1} and is assumed not to change due to the tuning, which is possible in thermal tuning. As shown in Fig. 8(a), with the change of the refractive index in DBR, the optical power calculated from the SS-TDM is nearly constant, while that calculated from the CWS-ICF decreases monotonically. The abnormal decrease of the output power occurs because, in the case of CWS-ICF, the transmission ($|A_{11}|$) through and reflection ($|A_{12}|$) from a subsection (Δz) deviate from their true values as the detuning factor becomes larger, that is, the refractive index moves away from the value of 3.283, which is illustrated in Fig. 8(b). On the contrary, the SS-TDM shows constant reflection and transmission as the refractive index in the DBR changes, which is expected in real devices because the refractive index change doesn't involve any loss from the assumption. The abnormal decrease of the reflectivity calculated from the CWS-ICF results in an abnormal power decrease during the tuning of two-section DBR laser diodes, as shown in Fig. 8(a).

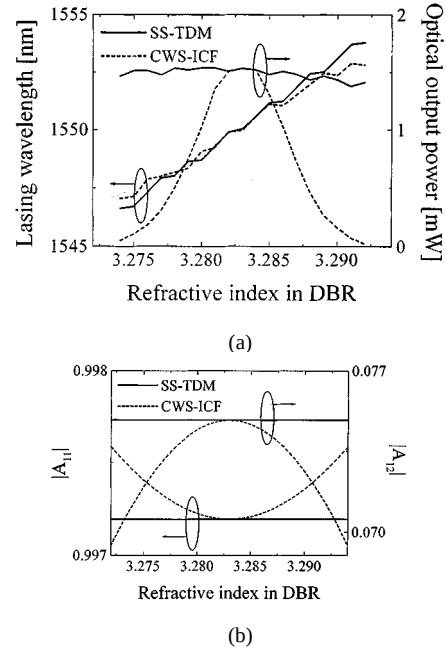


Fig. 8. Comparison of SS-TDM and CWS-ICF [18], when calculations are operated over the same two-section DBR laser diode. Effective phase refractive index without heating is assumed to be 3.283. (a) Lasing wavelength and optical output power as a function of the refractive index in DBR. (b) Reflection from and transmission through a subsection (Δz) as a function of the refractive index in DBR.

IV. CONCLUSIONS

An SS-TDM, which is very useful for modeling DFB or DBR laser diodes, was proposed and described in detail. In the SS-TDM, the time-dependent coupled wave equations are split into two sets of equations through the operator splitting concept. One set of the equations involves the time derivatives and the detuning factors. The other involves only the coupling terms. The numerical solutions of two sets of equations are found successively. Even though the error from the splitting is expected, it is found that accurate solutions can be obtained with quite a small number of subsections. The accuracy and efficiency of the SS-TDM are compared with other models, which are derived from the same coupled wave equations but with different numerical approaches.

The SS-TDM is found to require comparable computation time to the previous methods for the same number of subsections. However, in the SS-TDM, the accurate numerical solution can be obtained with an order of magnitude smaller number of subsections, which indicates the efficiency of the SS-TDM. The advantage of the SS-TDM is emphasized through the analysis of the complex-coupled DFB lasers and two-section DBR laser diodes.

In terms of the convergence of the output power from the complex-coupled DFB lasers, the SS-TDM and CWS-ICF shows the comparable performance, while they are greatly superior to the simple finite difference scheme. As the detuning factor increases due to the tuning, the reflectivity from and transmission through each subsection calculated using the CWS-ICF show erroneous results while those calculated

using the SS-TDM are reasonably accurate. The erroneous calculation of reflection and transmission in each subsection results in the abnormal decrease of power when the CWS-ICF is used. In the SS-TDM, the output power of the two-section DBR laser diode is maintained nearly constant during tuning. It is believed that the new SS-TDM can be widely used for the analysis and modeling of various semiconductor laser diodes and can also be adopted as an efficient source module for optical communication link simulators.

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